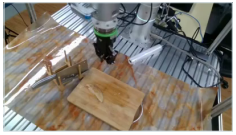
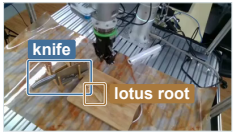
Action-aligned data construction

Task: Slice the lotus root with a knife.

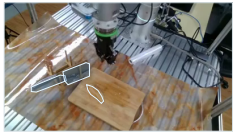
1 Input RGB · state · actions



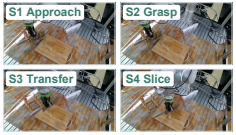
2 Detect GroundingDINO-B · Swin-B



3 Object masks SAM 3.1



4 Segment DINOv3 ViT-L/16



Phase · object / target · interaction grade

5 Segment record Mid

Approach the knife

Target: knife · Object box

Action window + state A T V

6 Grade-conditioned rewrites Qwen3.8-27B

- A Move downward.
- T Approach the knife.
- V Approach the object in the red box.

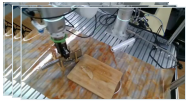
7 Review GPT-5.6-luna

- ✓ Semantic consistency
- ✓ Visual consistency
- ✓ Action alignment

Instruction examples

A Atomic motion T Task rewrite V Visual prompt

S1 Approach — Mid



A Move downward.

T Approach the knife.



V Approach the object in the red box.

S2 Grasp — High



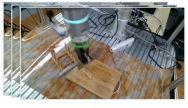
A N/A

T Grasp the knife.



V Grasp the object in the red box.

S3 Transfer / align — Mid



A Keep the gripper closed and move right.

T Align the knife above the root.



V Move the object toward the red point.

S4 Slice — High



A N/A

T Slice the lotus root.



V Slice the object in the red region.

S5 Withdraw — Mid



A Keep the gripper closed and move left.

T Withdraw the held knife.



V Move the object away from the red box.